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Research Paper

Development and validation of a stroke risk prediction model using regional healthcare big data and machine learning

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ABSTRACT

Objectives: This study aimed to develop and validate a stroke risk prediction model based on machine learning (ML) and regional healthcare big data, and determine whether it may improve the prediction performance compared with the conventional Logistic Regression (LR) model.

Methods: This retrospective cohort study analyzed data from the CHinese Electronic health Records Research in Yinzhou (CHERRY) (2015–2021). We included adults aged 18–75 from the platform who had established records before 2015. Individuals with pre-existing stroke, key data absence, or excessive missingness (>30 %) were excluded. Data on demographic, clinical measures, lifestyle factors, comorbidities, and family history of stroke were collected. Variable selection was performed in two stages: an initial screening via univariate analysis, followed by a prioritization of variables based on clinical relevance and actionability, with a focus on those that are modifiable. Stroke prediction models were developed using LR and four ML algorithms: Decision Tree (DT), Random Forest (RF), eXtreme Gradient Boosting (XGBoost), and Back Propagation Neural Network (BPNN). The dataset was split 7:3 for training and validation sets. Performance was assessed using receiver operating characteristic (ROC) curves, calibration, and confusion matrices, and the cutoff value was determined by Youden's index to classify risk groups.

Results: The study cohort comprised 92,172 participants with 436 incident stroke cases (incidence rate: 474/100,000 person-years). Ultimately, 13 predictor variables were included. RF achieved the highest accuracy (0.935), precision (0.923), sensitivity (recall: 0.947), and F1 score (0.935). Model evaluation demonstrated superior predictive performance of ML algorithms over conventional LR, with training/validation area under the curve (AUC)s of 0.777/0.779 (LR), 0.921/0.918 (BPNN), 0.988/0.980 (RF), 0.980/0.955 (DT), and 0.962/0.958 (XGBoost). Calibration analysis revealed a better fit for DT, LR and BPNN compared to RF and XGBoost model. Based on the optimal performance of the RF model, the ranking of factors in descending order of importance was: hypertension, age, diabetes, systolic blood pressure, waist, high-density lipoprotein Cholesterol, fasting blood glucose, physical activity, BMI, low-density lipoprotein cholesterol, total cholesterol, dietary habits, and family history of stroke. Using Youden's index as the optimal cutoff, the RF model stratified individuals into high-risk (>0.789) and low-risk ($\leq$0.789) groups with robust discrimination.

Conclusions: The ML-based prediction models demonstrated superior performance metrics compared to conventional LR and the RF is the optimal prediction model, providing an effective tool for risk stratification in primary stroke prevention in community settings.

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What is known?

- Stroke remains a leading cause of global mortality and long-term disability, highlighting the critical need for practical risk stratification tools.
- While several stroke risk prediction models exist, their performance may be limited when applied to diverse populations and healthcare settings, partly due to varying risk factor profiles.
- Machine learning (ML) algorithms have demonstrated superior capability in handling complex, nonlinear relationships in large datasets and show promise for improving predictive accuracy in healthcare over conventional statistical models.
- Nurses play a pivotal role in stroke primary prevention, but they lack a standardized tool for efficient risk screening.

What is new?

- The ML-based models can accurately predict the occurrence of stroke and outperform the conventional Logistic Regression (LR) method. Among the evaluated algorithms, the Random Forest (RF) demonstrated optimal predictive performance across all metrics.
- We developed and validated a novel stroke risk model for a community-based Chinese population. By focusing on modifiable risk factors, nurses can identify individuals at high risk and guide personalized interventions aimed at addressing unhealthy lifestyles.

1. Introduction

Stroke, clinically defined as an acute cerebrovascular event, represents a primary non-communacable chronic disease characterized by substantial incidence, mortality, recurrence rates, and long-term disability burden. According to the epidemiological survey data of the World Health Organization (WHO), stroke has become the second leading cause of death and the third leading cause of disability globally, with the number of deaths accounting for approximately 11 % of the total global deaths [1]. According to the Global Burden of Disease (GBD) Study 2019 [2], stroke accounted for 12.2 million incident cases, 101 million prevalent cases, and 6.55 million deaths globally, posing substantial challenges to public health security. Notably, the long-term functional impairments associated with stroke have created significant economic pressures on healthcare systems worldwide, particularly in China, due to the combined demands of acute care, rehabilitation, and chronic disease management.

Stroke risk factors comprise a complex interplay of non-modifiable (age, gender, low birth weight, ethnicity, and genetic predisposition) and modifiable determinants [3,4]. Population-attributable fraction analyses indicate that modifiable factors, including hypertension, diabetes, obesity, and lifestyle behaviors (smoking, alcohol use, physical inactivity, and unhealthy diet), account for 87.9 % of ischemic stroke and 89.5 % of hemorrhagic stroke disability-adjusted life years (DALYs) [2]. These findings underscore the critical importance of targeted primary prevention strategies addressing modifiable risk factors to reduce the global stroke burden [5]. Nevertheless, as revealed by the International Stroke Status Survey carried out by the World Stroke Organization (WSO) [6], only one-third of the primary stroke prevention strategies have been implemented in the surveyed countries. Notably, 41 % of these countries have either not implemented or only partially implemented such a strategy, indicating that primary stroke prevention strategies have not yet been widely adopted [6].

Risk stratification through stroke prediction models plays a pivotal role in targeted prevention strategies [7]. Globally, established risk assessment tools include the Framingham Stroke Risk Profile (FSRP) [8], QStroke model derived from the UK QResearch database [9], Pooled Cohort Equations (PCEs) [10], CHADS2 score for atrial fibrillation patients [11], and the PREDICT cardiovascular risk algorithm [12]. These validated instruments enable the identification of high-risk populations for the development of tailored interventions. However, risk prediction models developed from Western populations may not be directly generalizable to the Chinese population [13,14]. Studies have shown [14] that these models tend to overestimate stroke risk in Chinese cohorts, likely due to differences in population characteristics, risk factor profiles, and the underlying epidemiology of cardiovascular diseases. Currently, machine learning (ML), as one of the most widely utilized artificial-intelligence-driven health technologies, is frequently applied in assessing the morbidity risk of patients [7,15]. ML can integrate multi-source data such as demographic characteristics, clinical measures, and lifestyle factors to construct risk prediction models, providing crucial decision-making support for the primary prevention, early monitoring, and precise intervention of stroke. In recent years, several ML models have emerged in stroke research, primarily for predicting stroke prognosis [16–19] and various complications [20,21]. In contrast, the development of models to predict the risk of stroke development, particularly those emphasizing modifiable risk factors, has received comparatively less attention.

Nurses, positioned at the forefront of primary care and chronic disease management, play a pivotal role in stroke prevention. They are responsible for risk assessment, patient education, and coordinating long-term follow-up care. However, the effective execution of these duties is often hampered by the lack of efficient and precise risk stratification tools. Compounding this issue is the fact that relatively few stroke risk prediction models targeting the general population focus on modifiable lifestyle and behavioral patterns, which are crucial for nursing interventions [22]. Consequently, there is a pressing need to develop a predictive model that is not only statistically robust and clinically actionable but also incorporates these key modifiable factors. Although ML algorithms show promise for capturing complex, non-linear relationships, their “black-box” nature often hinders clinical interpretability and adoption [18]. In contrast, Logistic Regression (LR) remains a widely accepted and transparent benchmark in medical research due to its simplicity and ease of interpretation. To rigorously evaluate the added value of ML in this context, a comparative analysis against established LR models is essential. Therefore, this study aimed to develop and validate a stroke risk prediction model based on ML and regional healthcare big data, and determine whether it may improve the prediction performance compared with the conventional LR model. By facilitating early identification of high-risk individuals based on a comprehensive risk profile, this tool would empower community nurses to prioritize outreach, tailor health education, and implement targeted interventions proactively.

2. Methods

2.1. Study design and setting

The present study employed a retrospective cohort design for model development. It was reported in accordance with the Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis - Artificial Intelligence (TRIPOD-AI) [23] and the Strengthening the Reporting of Observational

Studies in Epidemiology (STROBE) [24] guidelines.

All the data used in this study are sourced from the CHinese Electronic health Records Research in Yinzhou (CHERRY) [25] dataset. Since 2009, this regional system has encompassed nearly all health-related activities of residents within this region, from birth to death, including children, adolescents, pregnant women, adults, and the elderly. Now, 98 % of permanent residents in Yinzhou have registered in the health information system. This study utilized data from multiple forms within the health record system, including basic personal health information, essential patient health data, chronic disease reports (e.g., patient registry cards), and health examination records. These distinct forms were linked via a de-identified unique patient identifier. The CHERRY study constitutes a large, population-based cohort that integrates individual-level electronic health records within a unique regional health information system. Its principal strength lies in the comprehensive capture of community-level data, particularly on cardiovascular risk factors and management from primary care, establishing it as a vital platform for clinical and epidemiological research in China. The data for this study covers the period from January 1, 2015, to October 31, 2021. The study cohort was restricted to participants enrolled after 2015 based on three primary considerations. First, data collected during the initial establishment of the database around 2009 may lack the robustness and completeness of later years. Second, more recent data better reflect contemporary temporal trends in risk factors. Third, this restriction ensured computational feasibility by managing dataset size, thereby mitigating potential constraints on memory and server capacity within the remote analysis platform.

2.2. Study participants and sample size

Patients meeting the following criteria are eligible for inclusion in this study: 1) individuals who had established records in the Yinzhou regional health management platform before January 1, 2015; 2) individuals whose record-keeping regions were not subject to administrative adjustments; and 3) individuals aged between 18 and 75 years. The exclusion criteria were: 1) incomplete identity or repeatedly records of the study subjects; 2) areas other than Yinzhou district; 3) absence of health examination information; 4) health examination information obtained after a confirmed diagnosis of stroke; 5) records indicating a diagnosis of stroke before January 1, 2015, as documented in the health records, health examination data, or chronic disease reports; and 6) variables or participants with a missing rate greater than 30 %.

The sample size for this study was determined in accordance with established standards for clinical prediction model development, which prioritize the number of outcome events (stroke) [26]. Informed by prior studies and clinical practice, we expected to include over 30 variables in the initial dataset. Based on the common rule of thumb requiring a minimum of 10 events per predictor variable (EPV), and leveraging the million-scale population registry of the CHERRY platform, our cohort provided a substantial number of eligible participants. This ensured that even after a 7:3 split into training and validation sets, the data volume remained sufficient to robustly develop and validate the model using both conventional LR and the four ML algorithms, thereby guaranteeing model stability and reliability.

2.3. Variables

Based on established literature [2,4,27] and clinical expertise, leveraging data availability from CHERRY, this study included variables across five aspects: 1) demographic characteristics: age, gender, weight, height, BMI, waist, marital status, occupational status, educational level, urban/rural, residential status; 2) clinical measures: systolic blood pressure (SBP), diastolic blood pressure (DBP), red blood cell counts (RBC), white blood cell counts (WBC), platelet count (PLT), fasting blood glucose (FBG), alanine aminotransferase (ALT), aspartate aminotransferase (AST), total bilirubin (TBIL), serum creatinine (CR), blood urea nitrogen (BUN), total cholesterol (TC), triglycerides (TG), low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein Cholesterol (HDL-C), albumin (ALB), direct bilirubin (DBIL); 3) lifestyle factors: dietary habits, physical activity, smoking; 4) comorbidities: hypertension, diabetes; and 5) family history of stroke. Dietary patterns were categorized according to the *Ningbo Smart Health Standard - Value Domain Code* into six types: balanced omnivore, meat-based, plant-based, preference for salty foods, preference for oily/fatty foods, and preference for sugary foods. For analytical purposes, dietary habits were dichotomized into “unhealthy dietary habits” and “healthy dietary habits.” “Unhealthy dietary habits” was defined as the presence of at least one preference for salty, oily, or sugary foods. “Healthy dietary habits” was defined as having a balanced omnivore pattern in the absence of any of these three unhealthy preferences. Participants were categorized based on their physical activity habits. Those who engaged in exercise less than twice per month or for less than 20 min per session were classified as having a poor exercise habit. Conversely, those who exercised at least twice a month, with each session lasting 20–40 min or longer, were considered to have a good exercise habit.

2.4. Study outcome

The primary outcome of this study was defined as the occurrence of stroke, including both hemorrhagic and ischemic strokes, as classified by the International Classification of Diseases, Tenth Revision (ICD-10) codes I60-I67 and I69. Stroke incidence was ascertained through patient registries and chronic disease records within the personal health archives. Individual records across different data sources were linked using a de-identified unique participant identifier. Follow-up continued until the first occurrence of the outcome event or the end of the study period (October 31, 2021), whichever came first.

2.5. Quality control

To safeguard the study’s validity, the entire process—including data preprocessing, variable definition, model development, and validation strategies—was meticulously guided and supervised by a team of two artificial intelligence experts and two methodologies. Every key step was iteratively discussed and confirmed by this expert panel before implementation, thereby guaranteeing the overall quality and credibility of our research.

2.6. Data analysis

2.6.1. General data analysis

All statistical analyses were performed with R (version 4.2.2) and Python (version 3.8.10). The count data were reported as frequency and percentage. Normally distributed continuous variables were expressed as mean and standard deviation, and non-normally distributed continuous variables were expressed as median (interquartile range). The count data were analyzed by the Pearson Chi-square (χ^2) test. If $1\leq$ theoretical frequency <5, the χ^2 correction of continuity (χ^2_c) was used. If the theoretical frequency <1, Fisher’s exact test was used for comparison. The continuous variables with equal variance were compared between groups using a *t*-test, and the constant variables with unequal variance

were analyzed using the Wilcoxon signed-rank test (ΣR). For all statistics, a two-sided *P*-value of <0.05 was considered statistically significant.

For those with a missing rate of less than 30 %, missing values were imputed. Continuous variables were imputed using the mean value, whereas categorical variables were imputed through multiple imputation techniques. Variable selection was performed using univariate analysis to assess the associations between independent variables and the outcome. Variables demonstrating a *P*-value <0.05 in the univariate analysis were initially selected, with an emphasis on identifying and including modifiable risk factors in the final model. We deliberately refrained from further multivariable analysis for variable selection, as our study primarily leveraged ML approaches for model development. ML algorithms are inherently capable of handling multiple predictors and capturing complex, nonlinear relationships. Preliminary multivariable filtering risked "premature information loss," potentially excluding variables that contribute meaningfully in ML-based models despite lacking significance in conventional linear models. Furthermore, variable inclusion was guided not solely by statistical thresholds but by a balanced consideration of clinical relevance and actionability, thereby enhancing the model's potential for future clinical translation in stroke prevention.

Given the highly possible imbalanced nature of the dataset for predicting stroke incidence, this study employed a hybrid sampling strategy to balance the sample sizes of stroke and non-stroke cases, taking into account data volume, computational resources, and training time. Random undersampling was applied to reduce the sample size of non-stroke cases. At the same time, the synthetic minority oversampling technique (SMOTE) [7,28] was employed to augment the sample size of stroke cases, thereby ensuring a balanced dataset for analysis.

2.6.2. Model development and comparison

This study employed a conventional LR model and four ML algorithms—Decision Tree (DT), Random Forest (RF), eXtreme Gradient Boosting (XGBoost), and Back Propagation Neural Network (BPNN)—to construct predictive models for stroke risk. The predictive performance of these models was compared. The balanced dataset was randomly split into training and validation sets at a ratio of 7:3. The training set was used for model development and training. In contrast, the validation set was utilized for model evaluation.

Model classification accuracy was assessed using a confusion matrix, which visually represents the discrepancies between predicted and actual outcomes. From the confusion matrix, metrics such as accuracy, precision, sensitivity, and F1 score were derived. The discriminative ability of the stroke prediction models was evaluated using the area under the curve (AUC), with a higher AUC indicating superior classification performance. The AUC value ranges from 0 to 1, with a value of 0.5 indicating no discriminative ability. An AUC between 0.5–0.7 is considered low discrimination, 0.7–0.9 moderate discrimination, and above 0.9 high discrimination. Calibration assesses the agreement between predicted probabilities and the observed frequencies of outcomes. Calibration was evaluated with calibration plots, which compare the predicted probabilities (x-axis) to the observed event frequencies (y-axis). A model is considered well-calibrated if its curve closely aligns with the 45-degree reference line, which denotes perfect prediction. The overall experimental design is illustrated in Appendix A.

2.6.3. Cutoff values and feature importance ranking

To enable risk stratification, an optimal cut-off value for the prediction model was determined by identifying the point on the receiver operating characteristic (ROC) curve that maximized Youden's index (J = sensitivity + specificity - 1). This approach ensures a balanced trade-off between sensitivity and specificity. Furthermore, the best-performing model was utilized to rank the importance of predictors for incident stroke.

2.7. Ethical considerations

This study was conducted with the approval of the Biomedical Ethics Committee of Peking University (Approval No. IRB00001052-22108).

3. Results

3.1. Characteristics of the participants

A total of 92,172 participants were included in this study, among whom 436 incident cases of stroke were identified, yielding an incidence rate of 474 per 100,000 person-years. The flowchart of participant selection is presented in Appendix B. The enrolled participants had a mean age of 57.31 years, with a higher proportion of females (57.07 %) compared to males (42.93 %). A significant majority of the participants (88.36 %) had attained an educational level of junior high school or lower. Detailed demographic characteristics are presented in Appendix C.

3.2. Imbalanced data resampling

A total of 92,172 participants were enrolled in this study, with 436 stroke cases identified, indicating a highly imbalanced dataset. To balance the data, SMOTE was employed to augment the stroke cases to 10,000, and random undersampling was utilized to reduce the non-stroke cases to 10,000, resulting in a 1:1 balanced dataset.

3.3. Variable selection

Univariate analysis identified 25 variables with $P < 0.05$, including age, gender, weight, BMI, waist, marital status, education, urban/rural, SBP, WBC, PLT, FBG, TBIL, CR, BUN, TC, LDL-C, HDL-C, ALB, DBIL, dietary habits, physical activity, hypertension, diabetes, and family history of stroke (Appendix D). Modifiable risk factors are more actionable for stroke prevention than non-modifiable ones. This study emphasizes modifiable risk factors, particularly lifestyle and behavioral changes. Although age and stroke family history are non-modifiable, they are critical for early risk assessment and should not be excluded. Thus, the final model incorporated 13 factors: age, BMI, waist, SBP, FBG, TC, LDL-C, HDL-C, family history of stroke, hypertension, diabetes, dietary habits, and physical activity.

3.4. Model development and comparison

Confusion matrices for the five models (LR, DT, RF, XGBoost, and BPNN) were generated for both the training and validation sets (Table 1). RF achieved the highest accuracy (0.935), precision (0.923), sensitivity (recall: 0.947), and F1 score (0.935), demonstrating superior classification performance and robustness compared to the other models. Overall, ML-based stroke risk prediction models outperformed the LR model, with RF being the most effective algorithm. These results were consistently replicated in the validation set.

ROC curves were plotted, and AUC values were calculated to evaluate model discriminative ability. For the training set (Fig. 1a), the AUC values for LR, DT, RF, XGBoost, and BPNN were 0.777, 0.980, 0.988, 0.962, and 0.921, respectively. RF achieved the

Table 1
Confusion matrices for the five models in the training and validation sets.

Models	Training sets				Validation sets			
	Accuracy	Precision	Recall	F1 Score	Accuracy	Precision	Recall	F1 Score
LR	0.708	0.692	0.743	0.717	0.716	0.706	0.752	0.728
BPNN	0.824	0.856	0.778	0.815	0.821	0.855	0.779	0.815
XGBoost	0.882	0.882	0.880	0.883	0.878	0.881	0.877	0.876
DT	0.914	0.922	0.904	0.913	0.894	0.904	0.886	0.895
RF	0.935	0.923	0.947	0.935	0.919	0.912	0.929	0.921

Note: LR = Logistic Regression. BPNN = Back Propagation Neural Network. XGBoost = eXtreme Gradient Boosting. DT = Decision Tree. RF = Random Forest.

highest AUC, indicating the best discriminative performance. In the validation set (Fig. 1b), the AUC values were 0.779, 0.955, 0.980, 0.958, and 0.918, respectively, consistent with the results obtained in the training set. ML-based models outperformed LR, with RF being the most effective.

Calibration evaluates the consistency between predicted risks and observed outcomes. In the training set (Fig. 2a), the DT, BPNN, and LR models exhibited better alignment with the ideal curve, indicating superior calibration compared to the RF and XGBoost models. In the validation set (Fig. 2b), the BPNN, LR, and DT models again showed better calibration, consistent with the results from the training set.

3.5. Cutoff values and feature importance ranking

Among the five models developed, those based on ML algorithms demonstrated superior overall predictive performance compared to the conventional LR model. The RF algorithm achieved the highest discriminative ability and was therefore selected for determining the risk stratification cutoff. The optimal cutoff value of 0.789 was identified using Youden's index, which effectively balances sensitivity and specificity. Individuals with a predicted risk $\geq$0.789 were classified as high-risk, while those below this threshold were classified as low-risk. Based on the optimal performance of the RF model, feature importance was evaluated to identify key predictors of stroke risk. The ranking of factors in descending order of importance was: hypertension, age, diabetes, SBP, waist, HDL-C, FBG, physical activity, BMI, LDL-C, TC, dietary habits, and family history of stroke (Appendix E).

4. Discussion

The 13 predictors ultimately included in this study collectively influence stroke risk through multiple interconnected pathophysiological pathways. Hypertension, diabetes, dyslipidemia (characterized by elevated TC, high LDL-C, and low HDL-C), and central obesity (reflected by waist circumference) constitute the core components of metabolic syndrome [29–31]. These factors significantly increase the risk of ischemic stroke by promoting atherosclerosis, endothelial dysfunction, and systemic inflammation [31]. Physical inactivity and unhealthy dietary habits further exacerbate obesity, insulin resistance, and metabolic disorders, creating a positive feedback loop [30]. Additionally, advanced age is a risk factor for arterial stiffening and diminished repair capacity [32]. At the same time, a family history of stroke may reflect both genetic susceptibility and cumulative risks derived from shared environmental exposures [33]. These factors are intertwined, collectively impairing vascular structure and hemodynamic stability, and ultimately leading to the occurrence of stroke events.

The superior performance of ML models over conventional LR observed in this study highlights ML's distinct advantages in modeling complex clinical data. Unlike LR, which is constrained by linear assumptions, ensemble methods effectively capture nonlinear relationships and interactions among risk factors [34]. Stroke pathogenesis involves multifaceted interactions between physiological, lifestyle, and clinical factors [2,4], which are unlikely to be fully modeled by simple linear assumptions. Furthermore, ML algorithms demonstrate stronger generalization capabilities in high-dimensional data settings through built-in feature selection and regularization, reducing the risk of overfitting. The robustness of RF against overfitting and its ability to handle a mix of continuous and categorical variables make it particularly suited for analyzing real-world, high-dimensional healthcare data [35]. Although the RF model exhibited less-than-ideal calibration, its significance is underscored by its primary strength: exceptional discriminative performance. This discriminative power, coupled with its stability and ability to manage high-dimensional data, is often paramount for clinical risk stratification [28], where

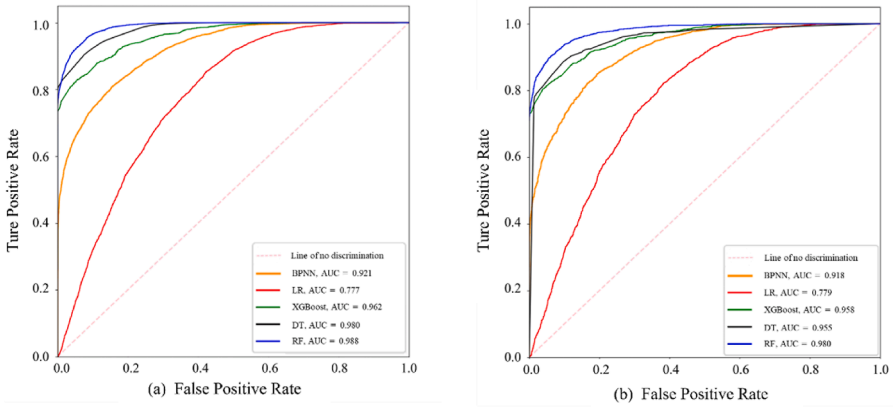


Fig. 1. The area under the curve values of the five models. (a) training set; (b) validation set. BPNN = Back Propagation Neural Network, LR = Logistic Regression, XGBoost = eXtreme Gradient Boosting, DT = Decision Tree, RF = Random Forest.

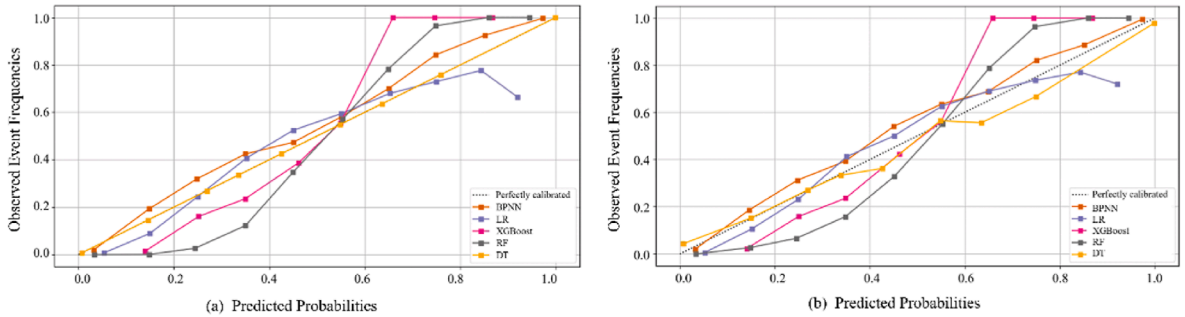


Fig. 2. Calibration curves of the five models. (a) training set; (b) validation set. BPNN = Back Propagation Neural Network, LR = Logistic Regression, XGBoost = eXtreme Gradient Boosting, RF = Random Forest, DT = Decision Tree.

accurately ranking patients by risk outweighs the need for perfect probability estimation. Therefore, future investigations should tailor the selection of prediction modeling methods to the specific characteristics of the dataset and clinical objectives, thereby enhancing their practical utility and generalizability across diverse healthcare settings.

The variable importance ranking derived from our model—hypertension, age, diabetes, SBP, waist, HDL-C, FBG, physical activity, BMI, LDL-C, TC, dietary habits, and family history of stroke—reflects both established pathophysiology and data-driven insights. The prominence of hypertension, SBP, and age aligns with their well-established roles as dominant drivers of endothelial dysfunction, arterial stiffness, and hemodynamic stress [31]. The high ranking of diabetes, FBG, and waist underscores the critical contribution of metabolic dysregulation, insulin resistance, and central obesity to stroke risk [29,36], likely through accelerated atherosclerosis and systemic inflammation. Behavioral factors, such as physical activity and diet, while important, may exhibit lower relative importance due to limitations in measurement granularity or their effects being partially mediated through physiological metrics, like waist circumference and blood pressure. Similarly, family history may rank lower, as its influence is likely indirect, acting through inherited predispositions to intermediate phenotypes such as hypertension or dyslipidemia [33]. For nursing practice, this ranking provides a data-driven basis for prioritizing interventions, with a focus on blood pressure control, glycemic management, and reducing central obesity. By integrating Youden's index-derived risk stratification, nurses can proactively identify high-risk individuals, allocate resources efficiently, and deliver targeted behavioral and monitoring interventions. The Youden's index of 0.789 was calculated, providing a validated cut-off value for future clinical application. This tool is readily deployable in community health centers or wellness management clinics. By inputting an individual's 13 risk factors, the model generates a comprehensive risk score. Individuals scoring above the 0.789 threshold are identified as high-risk for stroke, signaling the need for prompt and targeted interventions. These may include dietary modifications, weight management, and increased physical activity, thereby enabling proactive stroke prevention and enhancing public health capabilities at the community level.

This study developed a stroke risk prediction tool for clinical nursing practice. By identifying and prioritizing modifiable risk factors, such as waist circumference, physical activity, and dietary habits, nurses can deliver personalized, evidence-based primary prevention interventions. The model can be integrated into routine clinical workflows, enabling efficient and early screening of high-risk individuals, thereby allowing targeted preventive measures during the critical pre-symptomatic window. This proactive prevention strategy demonstrates significant clinical and economic benefits—the modest investment in systematic risk stratification is largely offset by long-term savings in stroke treatment and rehabilitation. The findings underscore the value of incorporating data-driven risk assessment into primary care and highlight the importance of reorienting healthcare systems toward preventive medicine to improve population cerebrovascular health. Ultimately, this tool helps optimize nursing efficiency and resource allocation, improving stroke-related health outcomes at the population level through early intervention and personalized management.

5. Limitations

This study had several limitations. First, although the present stroke risk prediction model was derived from a single institutional dataset, which may limit its generalizability, our research consortium has established a nationwide network of clinical demonstration sites. Subsequent studies will employ multi-regional data to iteratively optimize the model, strengthening its external validity and translational potential. Secondly, the prediction model was developed and internally validated using regional healthcare data, but it lacks external validation in independent cohorts. While the model demonstrates high discriminative ability in the derivation cohort, its reliance on single-region data poses risks of overfitting and limited external validity. Given potential variations in stroke epidemiology and healthcare systems across regions, the model's generalizability requires further investigation through multi-center studies. Thirdly, to address the class imbalance between stroke and non-stroke cases, SMOTE was applied to generate synthetic samples. This approach enhanced the model's sensitivity to minority class patterns while mitigating the risks of overfitting associated with naive oversampling. However, SMOTE-generated samples may not fully reflect clinical reality, particularly when feature correlations are complex. External validation is warranted to assess whether the model's performance generalizes to imbalanced real-world datasets in the subsequent research. Fourth, although the ML model developed in this study demonstrated excellent discriminative performance in stroke risk prediction, several methodological limitations should be acknowledged. As a typical black-box system, the current model lacks sufficient interpretability of its decision-making process. While explainability analysis methods such as Shapley Additive exPlanations (SHAP) have been proven effective in elucidating the decision mechanisms of nonlinear models [37,38], the following constraints precluded SHAP analysis in the present study: 1) restrictions on cross-institutional data sharing due to medical privacy protection requirements; 2) computational resource limitations; and 3) real-time clinical prediction latency requirements.

6. Conclusions

This study successfully developed and validated a stroke risk prediction model by leveraging a large-scale regional health database and comparing multiple ML algorithms with conventional LR. We find that the ML approaches, particularly the RF algorithm, outperformed LR in predictive accuracy. The final RF model exhibited strong discriminative ability and was effectively utilized to establish a clinically applicable risk stratification threshold. Furthermore, our RF-based analysis ranked the predictive importance of all variables, revealing that hypertension, age, and diabetes were the three most significant factors. This provides a concrete evidence base for primary stroke prevention by pinpointing the key risk factors that should be prioritized in clinical assessment and nursing practice. This model can be further developed into a clinical decision support system in the future, enabling the identification of high-risk patients and prioritizing interventions. This holds profound implications for reducing the incidence and alleviating the overall disease burden of stroke.

Data availability statement

The data that support the findings of this study are available from CHinese Electronic health Records Research in Yinzhou (CHERRY), but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Data are however available from the authors upon reasonable request and with permission of CHERRY.

CRediT authorship contribution statement

Yunxia Duan: Methodology, Software, Validation, Formal analysis, Writing – original draft, Writing – review & editing, Data curation, Visualization. **Rui Wang:** Methodology, Software, Validation, Formal analysis, Investigation, Data curation. **Yumei Sun:** Conceptualization, Investigation, Writing – review & editing, Supervision, Project administration. **Wendi Zhu:** Validation, Investigation, Writing – original draft. **Yi Li:** Conceptualization, Methodology, Software, Validation, Formal analysis. **Na Yu:** Methodology, Software, Validation, Data curation. **Yu Zhu:** Resources, Supervision, Validation, Project administration. **Peng Shen:** Resources, Supervision, Validation, Project administration. **Hongyu Sun:** Conceptualization, Methodology, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors utilized machine learning techniques in the field of artificial intelligence to develop a predictive model for stroke risk assessment. Following the use of this tool, the authors thoroughly reviewed and edited the generated content as necessary. The authors assume full responsibility for all aspects of the published work.

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Declaration of competing interest

The authors have declared no conflict of interest.

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Appendices. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijnss.2025.10.011.

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